Exercise 5 – Binary Number System (Negative Numbers)

Unsigned Binary Numbers	**Signed Binary Numbers**
Represents positive numbers only. All bits are used to represent values.	MSB is used for sign (0 for positive and 1 for negative numbers).
8-bit unsigned number can represent up to 2^8 values from 0 to 255. $(0)_{10}$ to $(255)_{10}$ $(00000000)_2$ to $(11111111)_2$	So for an 8-bit signed number, range will be 2^7: positive: $(0)_{10}$ to $(127)_{10}$ $(\mathbf{0}0000000)_2$ to $(\mathbf{0}1111111)_2$ negative: $(-1)_{10}$ to $(-127)_2$ $(\mathbf{1}0000000)_2$ to $(\mathbf{1}1111111)_2$
Example **3-bit unsigned** numbers: $(000)_2$ => 0 $(001)_2$ => 1 $(010)_2$ => 2 $(011)_2$ => 3 $(100)_2$ => 4 $(101)_2$ => 5 $(110)_2$ => 6 $(111)_2$ => 7	Example **3-bit signed** numbers: $(000)_2$ => 0 $(001)_2$ => +1 $(010)_2$ => +2 $(011)_2$ => +3 $(100)_2$ => -4 $(101)_2$ => -3 $(110)_2$ => -2 $(111)_2$ => -1
Convert Unsigned binary number to decimal: $(110)_2 \Rightarrow 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$ $\Rightarrow 4 + 2 + 0 = 6$	Convert **signed** binary number to decimal: negate the MSB value $(\mathbf{1}10)_2 \Rightarrow -1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$ $\Rightarrow -4 + 2 + 0 = -2$

Question: For 4-bit signed numbers, convert following to decimal values.

$(1001)_2$ $(1101)_2$ $(0101)_2$ $(0011)_2$

Question: Convert from decimal to 4-bit signed numbers (using double dabble).

-6 -4 +5 -1 +3